



BRENT OZAR
UNLIMITED®

Lab 1: Solving PAGEIOLATCH

1.3 p1

Setting up the workload

1. If you've been playing around changing tables or indexes, restore your StackOverflow database.
2. Restart your SQL Server instance. (For sp_BlitzIndex)
3. Copy & run the setup script for Lab 1.
4. Start SQLQueryStress:
 1. File Explorer, \Labs, double-click SQLQueryStress.exe
 2. Click File, Open, \Labs\ServerLab1.json
 3. Set "Number of Threads" based on your CPUs
 4. Click Go

1.3 p2



Bigger servers need bigger loads

Set “Number of Threads” to be 150% of your cores.

The screenshot shows the SQLQueryStress application window. The left pane contains the text "1 EXEC usp_ServerLab1". The right pane contains various configuration and monitoring fields. A red arrow points to the "Number of Threads" field, which is currently set to 6. Other fields include "Number of Iterations" (100), "Delay between queries (ms)" (100), "CPU Seconds/Iteration (Avg)", "Actual Seconds/Iteration (Avg)", "Elapsed Time", "Iterations Completed", "Total Exceptions", and "Logical Reads/Iteration (Avg)".

Field	Value
Number of Iterations	100
Number of Threads	6
Delay between queries (ms)	100
CPU Seconds/Iteration (Avg)	
Actual Seconds/Iteration (Avg)	
Elapsed Time	
Iterations Completed	
Total Exceptions	
Logical Reads/Iteration (Avg)	

1.3 p3

Don't know how many cores you have? Right-click on the taskbar, click Task Manager, Performance tab:

Utilization	Speed	Base speed:	2.30 GHz
21%	2.30 GHz	Sockets:	1
Processes	Threads	Virtual processors:	4



# of cores you have	SQLQueryStress Number of Threads
4	6
6	9
8	12
10	15



What the setup script does

Creates some stored procedures

Drops all nonclustered indexes

Sets MAXDOP = 1

(because we're not ready to cover parallelism yet)



What you're going to see

In SQLQueryStress, "Iterations Completed" won't move very quickly. (Or at all.)

Take long samples of waits:

```
sp_BlitzFirst @Seconds = 60, @ExpertMode = 1
```

Investigate to see if indexes will help:

```
sp_BlitzIndex @GetAllDatabases = 1
```

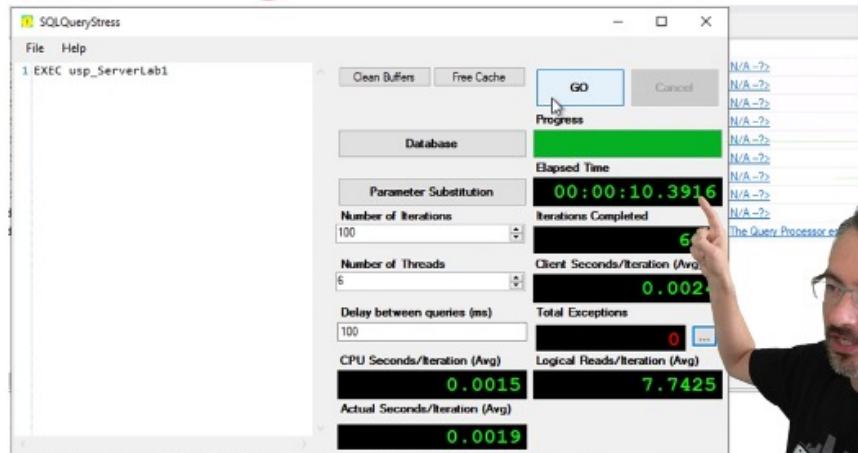
See what queries are causing those waits:

```
sp_BlitzCache @SortOrder = 'reads'
```

Add memory if you have to, but...remember, indexes/queries first. (Don't touch MAXDOP yet. More on that later.)



Your goal: a sub-60-second run.



1.3 p7



How I'd spend this 2 hours

1. Lunch (1 hour)
2. Check the server's top wait types, the queries causing those wait types, and write down your metrics (10 min)

```
sp_BlitzFirst @Seconds = 60, @ExpertMode = 1;  
sp_BlitzCache @SortOrder = 'reads'  
sp_BlitzIndex @GetAllDatabases = 1
```
3. Change a couple/few things on your server (5-10 min)
4. Go back to #2

1.3 p8



Stumped? Take a peek at Slack.

The Slack room lets you see:

- Other peoples' answers
- How quick (or slow) you are compared to them
- How thorough you are compared to them
- Who you should cheat off of during the final exam

1.3 p9

